JLSAE: Joint Low-Rank and Sparse Autoencoder for Hyperspectral Anomaly Detection

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***Abstract*—Without requiring prior spectral information, hyperspectral anomaly detection (HAD) relies on the identification of pixels that significantly deviate from the surroundings. Although autoencoders (AEs) are widely adopted, recent methods integrating low-rank and sparse priors often place these constraints in separate network branches, which hinders unified optimization and increases model complexity. To overcome these limitations, we propose a unified deep framework, termed joint low-rank and sparse AE (JLSAE), that embeds both priors within a single end-to-end architecture. The AE output serves as a learnable low-rank background, while an auxiliary sparse variable explicitly captures anomaly components. Crucially, we derive our optimization strategy from the alternating direction method of multipliers (ADMMs). This provides a principled training mechanism with explicit low-rank and sparsity regularization, thereby establishing strong physical interpretability by bridging deep learning with model-based optimization. Experimental results on multiple benchmark datasets show that JLSAE achieves more accurate background reconstruction and outperforms all twelve compared state-of-the-art methods in detection performance. The code is available at https://github.com/fan-jh/JLSAE**

***Index Terms*—Autoencoder (AE), background reconstruction, hyperspectral anomaly detection (HAD), low-rank prior, model-driven deep learning, sparse prior.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain abundant spectral details, allowing for the differentiation of diverse ground objects [1], [2]. By virtue of this advantage, researchers can locate targets of interest in a certain scene [3]. Target detection is broadly divided into two paradigms: supervised matched filtering, which assumes known target spectra, and unsupervised anomaly detection, which operates without prior spectral knowledge [4]. In practical applications, precise and comprehensive prior information about targets is usually not available [5], [6]. In such cases, hyperspectral anomaly detection (HAD) is widely adopted, as it does not rely on target spectral information but rather on the identification of pixels that significantly differ from the surroundings [5]. This technique has important applications in military reconnaissance, environmental monitoring, and mineral exploration [7].

Since anomalies are defined relative to the background, accurate background characterization is crucial for enhancing anomaly separability. Based on how the background is modeled, existing HAD approaches primarily fall into three categories: statistical modeling, structured reconstruction, and deep learning. Statistical modeling approaches assume that the background pixels are predominant and that their spectral characteristics follow a certain statistical distribution [8]. By estimating the distribution of the background, pixels that exhibit significant deviations from this distribution are identified as anomalous pixels. A representative example is the Global Reed-Xiaoli (GRX) detector [9], which specifies this certain statistical distribution as Gaussian. The GRX algorithm is simple but effective. Based on the GRX algorithm, many improved methods are further developed. The Local RX (LRX) algorithm [10] improves upon GRX by estimating the background statistics within a local spatial neighborhood around each test pixel, thereby adapting to nonstationary backgrounds. The weighted RX (WRX) method [11] improves anomaly detection performance by introducing a weighting scheme to better estimate background statistics and suppress interference from anomalous pixels. Moreover, the kernel-based RX (KRX) method [12] extends the conventional GRX detector into a high-dimensional feature space via the kernel trick, where background and anomalous pixels tend to be more distinctly separable, effectively capturing higher order correlations among spectral bands. Tao et al. [13] proposed a fractional Fourier entropy (FrFE)-based method to automatically determine the optimal intermediate domain for enhancing anomaly separability before statistical detection. Although these statistical-based methods have achieved promising results on certain datasets, their effectiveness in practical applications is often limited due to real-world backgrounds that do not conform to the distributional assumptions.

Received 11 June 2026; accepted 15 July 2026. Date of publication 23 July 2026; date of current version 31 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42325104, in part by China Postdoctoral Science Foundation under Grant 2026M790332, and in part by the Fundamental Research Funds for the Central Universities under Grant GK122601081. *(Corresponding author: Xiaotong Sun.)*

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Digital Object Identifier 10.1109/TGRS.2026.3716370

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In recent years, several studies have introduced topological structures into HAD to overcome the reliance of traditional statistical methods on assumptions about the background distribution [14]. However, such topological methods are computationally expensive and highly sensitive to parameter choices, which can hinder their practical applicability in large-scale or real-time scenarios.

In addition, structured reconstruction-based methods have also been well developed in recent years. The primary concept behind these methods is to reconstruct a clean background, and pixels showing larger reconstruction errors are more likely to be anomalies. Among these, collaborative representation (CR) models are grounded in the principle that background pixels can be linearly reconstructed from their spatial neighbors, while anomalous pixels fail to be accurately represented in this manner [15]. Beyond CR, researchers have also heavily investigated low-rank and sparse modeling techniques. This is motivated by two key properties of the hyperspectral data: spectral redundancy, which renders the background approximately low-rank, and anomaly rarity, which ensures their sparse occurrence across the scene [16]. The low-rank representation (LRR) method [17] harnesses the latent spectral structure of the background by inducing a rank-minimized approximation of the HSI to enable robust separation of anomalies. Building on LRR, the method in [18] imposes a sparsity constraint in the representation. Differently, the LRaSMD method in [19] decomposes the original image directly into the background and anomalies, thus eliminating the need for dictionary construction as in LRR, and employs the Go decomposition algorithm for solution optimization. In contrast to the global decomposition in [19], Cui et al. [20] proposed a local decomposition strategy where the image is first segmented into homogeneous superpixels using SLIC, and then a constrained LSD problem is solved within each superpixel. Furthermore, Zhang et al. [21] estimated the covariance matrix using the reconstructed clean background, thereby effectively mitigating the adverse effects of anomaly contamination when computing the Mahalanobis distance. Li et al. [22] employed a mixture-of-Gaussians prior on the sparse component to better capture the heterogeneity of complex anomalies and noise, and leveraged l_1-norm-based scoring on the residual for enhanced detection of subtle targets. Alternatively, the component decomposition analysis (CDA) model [23] orthogonally decomposes hyperspectral data into background, anomaly, and noise components via principal component analysis (PCA) and independent component analysis (ICA), leveraging both second-order and higher order statistics for separation. The effective anomaly space (EAS) concept [24] was proposed to further refine the anomaly component by incorporating sparsity cardinality constraints, thereby enhancing detectability through noise removal. In addition to the 2-D matrix method mentioned above, numerous 3-D tensor methods [25], [26], [27] have also emerged since 2022. The low-rank and sparse priors objectively exist within the HSIs. Hence, these construction methods are naturally interpretable. However, these methods typically require solving an energy minimization problem, with certain parameters needing manual adjustment (e.g., regularization parameters). Poor parameter selection can easily lead to local optima. In addition, the inherent heterogeneity and complexity of real-world HSI data make it challenging to choose a model that generalizes well across scenarios, which in turn degrades anomaly detection accuracy. Notably, Chang et al. [28] introduced an OSP-GoDec framework to autonomously determine the decomposition ranks without empirical tuning.

Deep learning has made great strides in recent years, particularly within computer vision, and is now being leveraged for HAD applications. Early deep learning-based methods were primarily built upon autoencoders (AEs) and generative adversarial networks (GANs) [29], [30], [31]. For instance, Wang et al. [29] proposed the Auto-AD network, which leverages a fully convolutional AE with skip connections and an adaptive-weighted loss function to autonomously reconstruct the background while suppressing anomaly reconstruction. Jiang et al. [31] proposed the HADGAN framework, which utilizes a latent discriminator and image feature corrector to guide the generator toward reconstructing the background while suppressing anomalies. In pursuit of more sophisticated feature representation, recent cutting-edge research has shifted toward advanced architectures beyond conventional convolutional networks. In particular, Transformer-based models have gained prominence due to their superior capability in modeling long-range dependencies and global context [32]. For example, Wang et al. [33] proposed the GIFormer to address granularity inconsistency in reconstruction. More recently, diffusion models have emerged as a powerful generative paradigm. Representatively, DWSDiff [34] introduces a dual-window spectral diffusion strategy to model the background distribution, demonstrating the potential of score-based generative models in HAD. Furthermore, the emergence of state-space models has introduced a new paradigm, with Mamba-based architectures like CWIMamba [35], ACMamba [36], and SSHAD [37] demonstrating remarkable efficiency in capturing multiscale spectral–spatial features through selective state spaces. Notably, despite the emergence of new architectures, the AE community remains vibrant and is actively evolving to address complex scenarios. Recent state-of-the-art works, such as the multiscale AE suppression strategy (MASS) [7], the dual-stream spatial–spectral fusion network (DSFNet) [38], and the multiscale memory network (MSMNet) [39], have demonstrated remarkable performance in 2025 and 2026. These advances indicate that the frontier of AE research is shifting toward more sophisticated designs that can better capture the intrinsic characteristics of hyperspectral data.

While the aforementioned works (e.g., MASS and DSFNet) focus on architectural complexity to enhance feature extraction, another prominent direction in recent AE research is the integration of physical priors. This approach seeks to improve interpretability by incorporating explicit mathematical constraints from traditional model-based methods into deep AE frameworks [40]. For example, DeepLR [41] introduces a low-rank prior for background modeling to guide the network training. Similarly, Feng et al. [42] combined an AE with a tensor-based background prior that enforces global low-rankness and local smoothness. Both [43] and [44] impose low-rank constraints on the latent representations learned by

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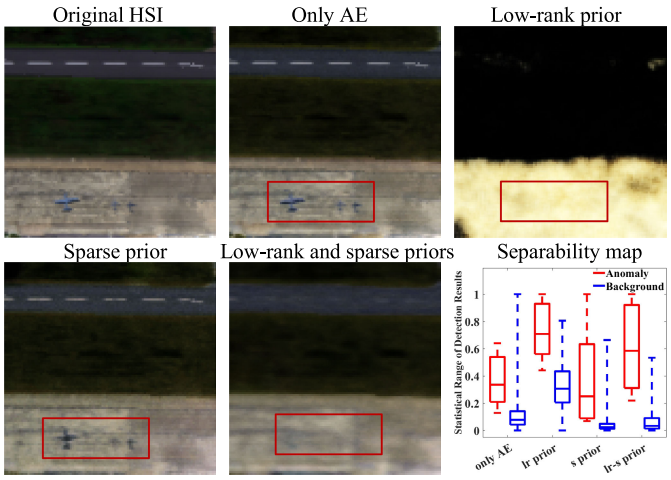


Fig. 1. Background reconstruction and separability map under different prior constraints.

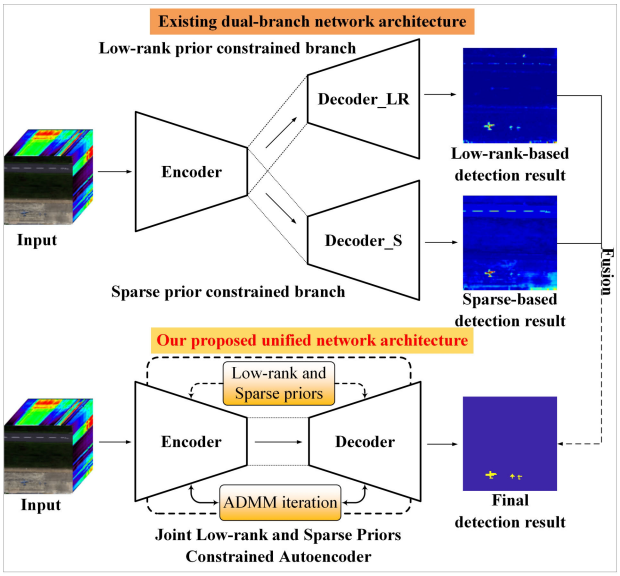


Fig. 2. Comparison of dual-branch and unified network architectures for HAD.

AEs to model the background as lying in a compact subspace. Lin et al. [45] recently introduced a dual-decoder architecture where one branch enforces a low-rank constraint for the background reconstruction and the other imposes a sparsity constraint for anomaly modeling. Hu et al. [46] adopted two AEs, imposing a low-rank prior on one and a sparse prior on the other to generate two separate anomaly maps, which were subsequently fused to produce the final detection result.

As can be seen above, integrating physical prior knowledge with deep AEs trained on the data has become a prevailing paradigm. This strategy not only mitigates the limited generalizability of traditional model-based methods but also enhances the interpretability of deep learning models. However, existing AE-based approaches predominantly focus on enforcing low-rank constraints to model the background [41], [42], [43], [44], largely overlooking the critical role of sparse priors. As illustrated in Fig. 1, when relying solely on an AE, the anomalies may also be reconstructed due to its powerful representation capability. While a low-rank prior alone effectively suppresses anomalies, it often leads to oversmoothing and loss of fine background details; conversely, a sparse prior may preserve textures but risks reconstructing the anomalies. On the contrary, jointly enforcing low-rank and sparse priors leads to significantly better background reconstruction and anomaly suppression than either prior alone, achieving the largest separation margin between background and anomalies in the separability map. Although a few recent works [45], [46] have attempted to incorporate both low-rank and sparse priors, they typically apply these constraints separately within two distinct network branches, as shown in Fig. 2. This dual-branch design assigns low-rank and sparse priors to two independent decoders, hindering their joint optimization and inevitably increasing model complexity. Moreover, the detection results from each branch require subsequent fusion to obtain the final detection result.

Considering the above issues, we introduce a deep low-rank and sparse priors learning framework, termed joint low-rank and sparse AE (JLSAE), which unifies both priors within a single, end-to-end trainable architecture via alternating direction method of multipliers (ADMM)-based optimization, as shown in the lower part of Fig. 2. By coupling low-rank and sparse regularization, our method achieves a better balance between background suppression and anomaly enhancement, leading to superior detection performance. The detailed flowchart is illustrated in Fig. 3. In particular, our model employs a purely convolutional AE with shortcut connections to preserve multiscale spatial details while reconstructing the background. To reconstruct a clean background while effectively suppressing anomalies, we propose an innovative model that tightly couples low-rank and sparse matrix decomposition with the AE architecture. Crucially, the network output is interpreted as the low-rank background, while anomalies are explicitly captured by a sparse auxiliary variable—preventing contamination of the reconstruction. This formulation is optimized via an ADMM-inspired learning scheme, which naturally yields a composite loss function comprising low-rank and sparsity regularization terms. Each training iteration corresponds to one ADMM step, bridging deep learning with model-based optimization. An adaptive weighting strategy further suppresses anomalous influence during training. The final detection map is generated from the reconstruction error. By unifying the low-rank and sparse priors within a single neural architecture with shared optimization, our method achieves both strong interpretability and high detection performance.

Overall, this article offers three primary contributions, which are listed as follows.

1) We propose a novel end-to-end unified deep architecture that embeds low-rank and sparse priors within a single AE, replacing the existing dual-branch paradigm. This unified design enables coherent joint optimization, eliminates the subsequent fusion step, and reduces model redundancy.
2) We develop a joint low-rank and sparse modeling framework for the HAD that simultaneously captures the global correlation of background pixels and the isolated nature of anomalies, effectively mitigating the

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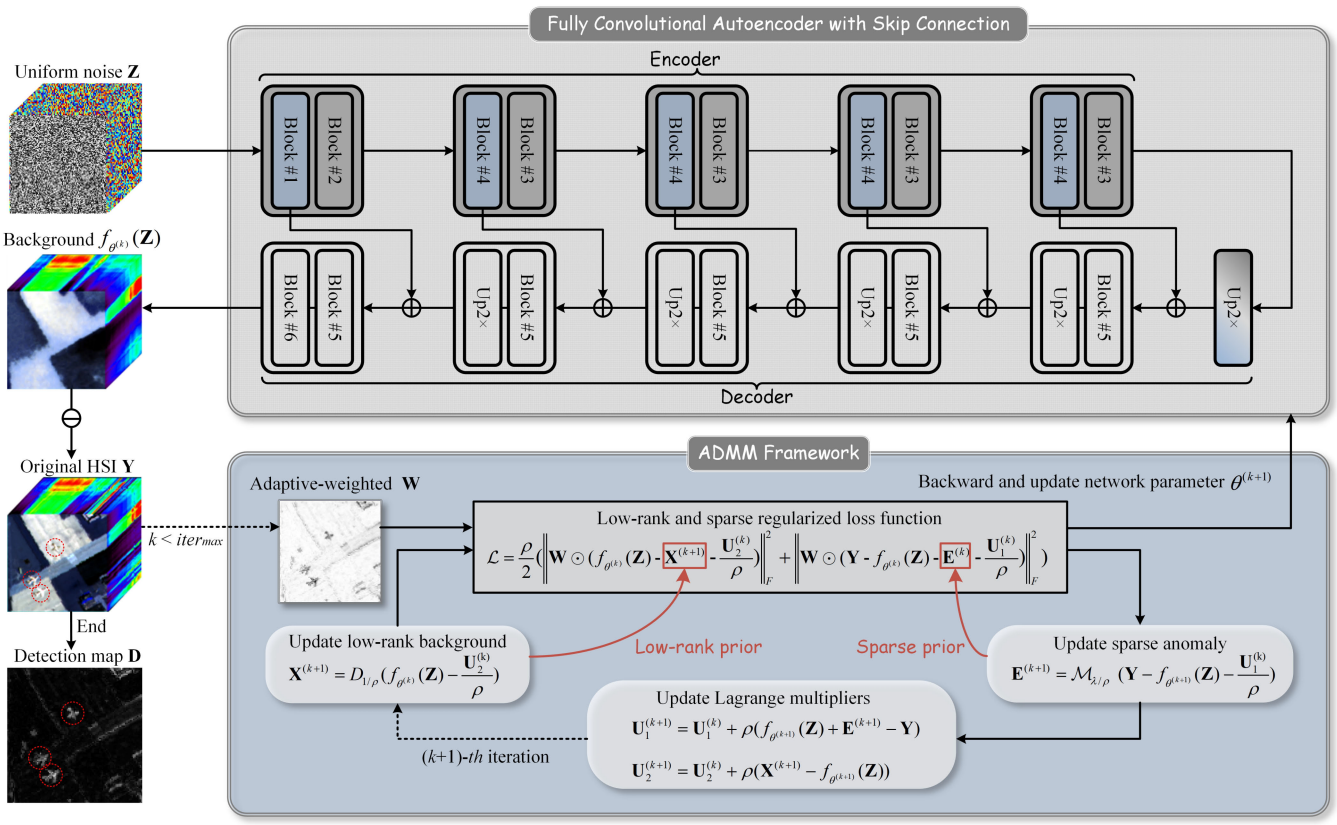


Fig. 3. Overview of the proposed JLSAE.

oversmoothed background reconstruction caused by low-rank-only deep methods.

3) We derive a physically interpretable loss function with explicit low-rank and sparsity regularization by interpreting each training iteration as an ADMM step, thereby establishing a principled connection between deep learning and model-based optimization and enhancing the interpretability of the proposed model.

II. Related Work

This section reviews the fundamentals of robust PCA (RPCA) and deep AEs, which form the basis of the proposed method.

A. Robust PCA

The purpose of RPCA is to decompose a given matrix $\mathbf{Y} \in \mathbb{R}^{m\times n}$ into a low rank matrix $\mathbf{X} \in \mathbb{R}^{m\times n}$ and sparse matrix $\mathbf{E} \in \mathbb{R}^{m\times n}$

$$\mathbf{Y} = \mathbf{X} + \mathbf{E} \tag{1}$$

where $\mathbf{X}$ and $\mathbf{E}$ are estimated via minimization of the following objective:

$$\min_{\mathbf{X},\mathbf{E}} \|\mathbf{X}\|_* + \lambda\|\mathbf{E}\|_1 \quad \text{s.t. } \mathbf{Y} = \mathbf{X} + \mathbf{E}. \tag{2}$$

Here, $\|\cdot\|_*$ represents the nuclear norm promoting a low-rankness, $\|\cdot\|_1$ denotes the l_1-norm enforcing sparsity, and λ balances the contributions of the two optimization objectives.

The model of RPCA has many important applications in video surveillance, face recognition, and collaborative filtering [47]. Notably, the background in HSIs is usually composed of a limited set of land cover types, whose spectral signatures linearly mix to form all background pixels, rendering the background data matrix approximately low-rank due to the inherent spectral redundancy. Furthermore, anomalous pixels typically constitute only a minimal fraction, making them sparse. Therefore, RPCA is well-suited for the HAD [48].

B. Autoencoder

The core operation of an AE involves a nonlinear transformation from the complex input space to a compressed latent manifold, followed by a reverse mapping back to the original space. In particular, given an input sample $\mathbf{y} \in \mathbb{R}^n$, the encoder computes its latent representation $\mathbf{z} \in \mathbb{R}^m$ $(m < n)$ as

$$\mathbf{z} = f_{\theta_e}(\mathbf{y}) \tag{3}$$

while the decoder attempts to recover the original signal

$$\tilde{\mathbf{y}} = f_{\theta_d}(\mathbf{z}). \tag{4}$$

Here, θ_e and θ_d denotes the learnable parameters associated with the encoder and decoder, respectively. These parameters are jointly optimized via gradient descent to reduce the discrepancy between the input $\mathbf{y}$ and its approximation $\tilde{\mathbf{y}}$, formalized as

$$\mathcal{L}_1 = \|\mathbf{y} - \tilde{\mathbf{y}}\|_2^2. \tag{5}$$

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The AEs have been widely adopted in image denoising [49], dimensionality reduction [50], [51], and anomaly detection in industrial inspection and medical imaging [52], [53]. In recent years, the AEs have also been introduced into HAD. Given that background pixels constitute the vast majority of the original image, the network tends to capture their dominant feature patterns and reconstructs the background pixels more easily than the anomalous pixels. This discrepancy leads to higher reconstruction errors for anomalous pixels, which can be directly used as the anomaly score.

III. Proposed JLSAE

We present the JLSAE, a single-branch, end-to-end AE that jointly embeds low-rank and sparse priors into its learning objective via an ADMM-inspired regularization scheme. The proposed framework, depicted in Fig. 3, comprises a convolutional AE and an ADMM solver. By solving the formulated problem via ADMM, an iterative scheme is obtained, which naturally yields a composite loss function comprising low-rank and sparsity regularization terms and thereby establishes the connection between the AE and the ADMM framework. Each network update corresponds to one iteration, encouraging the model to enforce the low-rank and sparse structures in its internal representations and guiding it to generate a clean background.

A. Problem Formulation

According to the theory of AE, the background of an HSI $\mathbf{Y} \in \mathbb{R}^{b\times(w\times h)}$ (with b spectral bands and spatial dimensions $w \times h$) can be reconstructed through the following self-similarity loss function:

$$\mathcal{L} = \|\mathbf{Y} - f_\theta(\mathbf{Z})\|_F^2 \tag{6}$$

where $\mathbf{Z} \in \mathbb{R}^{b\times(w\times h)}$ is a noise image that has the same dimensions as $\mathbf{Y}$, generated via independent random sampling from a uniform distribution. θ denotes the learnable parameters of the network and $\|\cdot\|_F$ denotes the Frobenius norm. Following [29] and [41], the architecture is fed with a random noise $\mathbf{Z}$ instead of the observed image $\mathbf{Y}$. The network is optimized to reconstruct $\mathbf{Y}$ from $\mathbf{Z}$, effectively learning a mapping that generates plausible background structures. By minimizing the loss function (6), the network output $f_\theta(\mathbf{Z})$ gradually converges to $\mathbf{Y}$. Given that the background pixels constitute the vast majority of the original image, AE reconstructs them more easily than anomalous pixels [29], and the reconstruction error serves as an indicator of anomalies.

Although anomaly detection can be achieved through the way mentioned above, it does not fully exploit the fact that the background exhibits a high spectral redundancy, while anomalies appear only at a few spatially isolated locations. Therefore, the anomalies may also be reconstructed due to the powerful representation capability of the AEs.

To fully exploit the prior knowledge associated with the hyperspectral background and anomalous targets, we formulate a minimization optimization problem that tightly couples RPCA with the AE architecture. In this formulation, the network output is interpreted as the low-rank background, while anomalies are explicitly captured by a sparse auxiliary variable, thereby preventing them from contaminating the reconstructed background

$$\begin{aligned} &\min_{\mathbf{E},\theta} \|f_\theta(\mathbf{Z})\|_* + \lambda \|\mathbf{E}\|_{2,1} \\ &s.t.\ \mathbf{Y} = f_\theta(\mathbf{Z}) + \mathbf{E} \end{aligned} \tag{7}$$

where $\mathbf{E} \in \mathbb{R}^{b\times(w\times h)}$ represents the anomaly matrix and $\|\cdot\|_{2,1}$ sums the l_2-norms of all matrix columns, encouraging as many zero columns in $\mathbf{E}$ as possible, thereby enforcing sparsity. $\|\cdot\|_*$ denotes the nuclear norm of a matrix promoting low-rankness. λ is a regularization term used to balance the contributions of the two optimization objectives.

By replacing the low-rank matrix $\mathbf{X}$ in (1) with the network output $f_\theta(\mathbf{Z})$, the AE is guided to generate a low-rank background representation. Simultaneously, the residual term $\mathbf{E} = \mathbf{Y} - f_\theta(\mathbf{Z})$ also contains the learnable parameter θ of the AE. Therefore, by minimizing the $l_{2,1}$-norm of $\mathbf{E}$, the AE is constrained by a sparse prior naturally. This allows for the effective reconstruction of a low-rank, clean background while simultaneously accounting for the sparsity of anomalies, thus preventing contamination of the reconstruction.

The solution procedure of (7) is detailed in Section III-B, and the detailed description of the network architecture is provided in Section III-C.

B. Optimization via ADMM

To avoid direct differentiation through the nonsmooth nuclear norm (associated with the low-rank term), we embed the model within an ADMM framework [54]. By introducing an auxiliary background variable $\mathbf{X} = f_\theta(\mathbf{Z})$ during optimization, we circumvent gradient computation on the nuclear norm altogether

$$\begin{aligned} &\min_{\theta,\mathbf{X},\mathbf{E}} \|\mathbf{X}\|_* + \lambda \|\mathbf{E}\|_{2,1} \\ &\text{s.t. } \mathbf{Y} = f_\theta(\mathbf{Z}) + \mathbf{E} \\ &\quad\ \mathbf{X} = f_\theta(\mathbf{Z}). \end{aligned} \tag{8}$$

To maintain notational consistency with the RPCA model in (1), we denote this auxiliary variable as $\mathbf{X}$, representing the reconstructed low-rank background component. The augmented Lagrangian method [55] is then employed to convert the equality-constrained terms into penalty terms within the Lagrangian function

$$\begin{aligned} &L_\rho(\mathbf{X}, \theta, \mathbf{E}, \mathbf{U}_1, \mathbf{U}_2) \\ &\quad = \|\mathbf{X}\|_* + \lambda \|\mathbf{E}\|_{2,1} \\ &\qquad + \frac{\rho}{2}\left(\|f_\theta(\mathbf{Z}) + \mathbf{E} - \mathbf{Y}\|_F^2 + \|\mathbf{X} - f_\theta(\mathbf{Z})\|_F^2 \right) \\ &\qquad + tr\left(\mathbf{U}_1^T (f_\theta(\mathbf{Z}) + \mathbf{E} - \mathbf{Y})\right) + tr\left(\mathbf{U}_2^T \left(\mathbf{X} - f_\theta(\mathbf{Z})\right)\right) \end{aligned} \tag{9}$$

where ρ is a penalty parameter influencing the convergence behavior, $\mathbf{U}_1, \mathbf{U}_2 \in \mathbb{R}^{b\times(w\times h)}$ represent the Lagrange multipliers, and tr$(\cdot)$ denotes the trace of a matrix. Equation (9) can be equivalently transformed into

$$L_\rho(\mathbf{X}, \theta, \mathbf{E}, \mathbf{U_1}, \mathbf{U_2}) = \|\mathbf{X}\|_* + \lambda \|\mathbf{E}\|_{2,1}$$

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$$+\frac{\rho}{2}\left(\left\|\mathbf{Y}-f_\theta(\mathbf{Z})-\mathbf{E}-\frac{\mathbf{U}_1}{\rho}\right\|_F^2+\left\|f_\theta(\mathbf{Z})-\mathbf{X}-\frac{\mathbf{U}_2}{\rho}\right\|_F^2\right)-\frac{1}{2\rho}\left(\|\mathbf{U}_1\|_F^2+\|\mathbf{U}_2\|_F^2\right). \tag{10}$$

The original optimization problem (8) is then addressed by minimizing the augmented Lagrangian function (10). With the ADMM framework, which is selected to serve as a vital bridge connecting the mathematical priors of the RPCA with the learnable nature of deep neural networks, this process involves iteratively updating the variables $\mathbf{X}$, θ, $\mathbf{E}$, $\mathbf{U}_1$, and $\mathbf{U}_2$, thereby decomposing the original optimization task into alternating, tractable subproblems that admit closed-form or efficient solutions. Starting from the initial variables $(\mathbf{X}^{(0)}, \theta^{(0)}, \mathbf{E},^{(0)} \mathbf{U}_1^{(0)}, \mathbf{U}_2^{(0)})$, the variables are updated at each iteration $k \geq 0$ according to

$$\begin{cases} X^{(k+1)} = \arg\min_X L_\rho(X, \theta^{(k)}, E^{(k)}, U_1^{(k)}, U_2^{(k)}) \\ \theta^{(k+1)} = \arg\min_\theta L_\rho(X^{(k+1)}, \theta, E^{(k)}, U_1^{(k)}, U_2^{(k)}) \\ E^{(k+1)} = \arg\min_E L_\rho(X^{(k+1)}, \theta^{(k+1)}, E, U_1^{(k)}, U_2^{(k)}) \\ \begin{cases} U_1^{(k+1)} = U_1^{(k)} + \rho(f_{\theta^{(k+1)}}(Z) + E^{(k+1)} - Y) \\ U_2^{(k+1)} = U_2^{(k)} + \rho(X^{(k+1)} - f_{\theta^{(k+1)}}(Z)). \end{cases} \end{cases} \tag{11}$$

In each step, with the other variables held constant, the background $\mathbf{X}$, network parameters θ, anomaly matrix $\mathbf{E}$ and Lagrange multipliers are updated sequentially. By substituting the Lagrangian function (10) into (11) and omitting terms independent of the optimization variables, (11) can be further expressed as

$$\begin{cases} \mathbf{X}^{(k+1)} = \arg\min_{\mathbf{X}} \|\mathbf{X}\|_* + \frac{\rho}{2}\left\|f_{\theta^{(k)}}(\mathbf{Z}) - \mathbf{X} - \frac{\mathbf{U}_2^{(k)}}{\rho}\right\|_F^2 \\ \theta^{(k+1)} = \arg\min_\theta \frac{\rho}{2}\left(\left\|\mathbf{Y} - f_\theta(\mathbf{Z}) - \mathbf{E}^{(k)} - \frac{\mathbf{U}_1^{(k)}}{\rho}\right\|_F^2 + \left\|f_\theta(\mathbf{Z}) - \mathbf{X}^{(k+1)} - \frac{\mathbf{U}_2^{(k)}}{\rho}\right\|_F^2\right) \\ \mathbf{E}^{(k+1)} = \arg\min_{\mathbf{E}} \lambda\|\mathbf{E}\|_{2,1} + \frac{\rho}{2}\left\|\mathbf{Y} - f_{\theta^{(k+1)}}(\mathbf{Z}) - \mathbf{E} - \frac{\mathbf{U}_1^{(k)}}{\rho}\right\|_F^2 \\ \begin{cases} \mathbf{U}_1^{(k+1)} = \mathbf{U}_1^{(k)} + \rho(f_{\theta^{(k+1)}}(\mathbf{Z}) + \mathbf{E}^{(k+1)} - \mathbf{Y}) \\ \mathbf{U}_2^{(k+1)} = \mathbf{U}_2^{(k)} + \rho(\mathbf{X}^{(k+1)} - f_{\theta^{(k+1)}}(\mathbf{Z})). \end{cases} \end{cases} \tag{12}$$

The solution methods for each subproblem are detailed below.

1) *To Update Background* $\mathbf{X}$: Fix $f_{\theta^{(k)}}(\mathbf{Z})$ and $\mathbf{U}_2^{(k)}$. Then $\mathbf{X}$ is updated using (13)–(15)

$$\mathbf{X}^{(k+1)} = D_{1/\rho}\left(f_{\theta^{(k)}}(\mathbf{Z}) - \frac{\mathbf{U}_2^{(k)}}{\rho}\right) \tag{13}$$

where $D_{1/\rho}(\cdot)$ denotes the singular value thresholding operator [56], which filters out all singular values smaller than $1/\rho$. In particular

$$D_{1/\rho}(\mathbf{G}) = \mathbf{U}S_{1/\rho}(\mathbf{\Sigma})\mathbf{V}^T \tag{14}$$

with $\mathbf{G} = f_{\theta^{(k)}}(\mathbf{Z}) - (\mathbf{U}_2^{(k)}/\rho)$, and $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ being the singular value decomposition of $\mathbf{G}$. Here, $S_{1/\rho}(\cdot)$ is the shrinkage operator applied element wise to the diagonal entries of $\mathbf{\Sigma}$, defined as

$$S_{1/\rho}(x) = sgn(x)\max\left(|x| - \frac{1}{\rho}, 0\right). \tag{15}$$

In this expression, $x \in \mathbb{R}$, sgn$(\cdot)$ is the sign function, and max$(\cdot)$ returns the larger of two arguments.

2) *To Update Network Parameter* θ: Fix $\mathbf{X}^{(k+1)}, \mathbf{E}^{(k)}, \mathbf{U}_1^{(k)}$ and $\mathbf{U}_2^{(k)}$. The subproblem for updating the network parameter θ is defined by

$$\theta^{(k+1)} = \arg\min_\theta \frac{\rho}{2}\left(\left\|\mathbf{Y} - f_\theta(\mathbf{Z}) - \mathbf{E}^{(k)} - \frac{\mathbf{U}_1^{(k)}}{\rho}\right\|_F^2 + \left\|f_\theta(\mathbf{Z}) - \mathbf{X}^{(k+1)} - \frac{\mathbf{U}_2^{(k)}}{\rho}\right\|_F^2\right). \tag{16}$$

Using (16) as the loss function, each network update corresponds to one ADMM iteration step, which bridges deep learning with model-based optimization. Compared to the original loss function in (6), the loss in (16) incorporates both the information of background $\mathbf{X}^{(k+1)}$ and the information of anomalies $\mathbf{E}^{(k)}$. In other words, this ADMM-based derivation naturally yields a composite loss function consisting of two terms: 1) a low-rank regularization term that enforces the global spectral correlation in the reconstruction and 2) a sparsity-promoting term that explicitly penalizes the network from fitting anomalous structures. The detailed network architecture and adaptive weighting mechanism for the loss function are further discussed in Section III-C.

3) *To Update Anomaly Matrix* $\mathbf{E}$: Fix $f_{\theta^{(k+1)}}(\mathbf{Z})$ and $\mathbf{U}_1^{(k)}$. The anomaly matrix $\mathbf{E}$ is updated using (17) and (18)

$$\mathbf{E}^{(k+1)} = \mathcal{M}_{\lambda/\rho}\left(\mathbf{Y} - f_{\theta^{(k+1)}}(\mathbf{Z}) - \frac{\mathbf{U}_1^{(k)}}{\rho}\right) \tag{17}$$

where $\mathcal{M}_{\lambda/\rho}(\cdot)$ is the soft thresholding operator [57]. $\mathcal{M}_{\lambda/\rho}(\cdot)$ sets each column of the input matrix whose l_2-norm is less than λ/ρ to zero, and scales each column whose l_2-norm is larger than λ/ρ. In particular, the quantity is derived as

$$\mathbf{e}_i^{(k+1)} = \frac{\mathbf{h}_i}{\|\mathbf{h}_i\|_2}\max\left(\|\mathbf{h}_i\|_2 - \frac{\lambda}{\rho}, 0\right) \tag{18}$$

where $\mathbf{e}_i^{(k+1)}$ corresponds to the ith column vector of matrix $\mathbf{E}^{(k+1)}$, and $\mathbf{h}_i$ is the ith column vector of $\mathbf{H} = \mathbf{Y} - f_{\theta^{(k+1)}}(\mathbf{Z}) - (\mathbf{U}_1^{(k)}/\rho)$.

4) *To Update Lagrange Multipliers* $\mathbf{U}$: Fix $\mathbf{X}^{(k+1)}, \mathbf{E}^{(k+1)}$, and $f_{\theta^{(k+1)}}(\mathbf{Z})$. The Lagrange multipliers can be updated using the following equation:

$$\begin{cases} \mathbf{U}_1^{(k+1)} = \mathbf{U}_1^{(k)} + \rho(f_{\theta^{(k+1)}}(\mathbf{Z}) + \mathbf{E}^{(k+1)} - \mathbf{Y}) \\ \mathbf{U}_2^{(k+1)} = \mathbf{U}_2^{(k)} + \rho(\mathbf{X}^{(k+1)} - f_{\theta^{(k+1)}}(\mathbf{Z})). \end{cases} \tag{19}$$

At this stage, the entire iterative procedure for solving (8) via ADMM has been thoroughly described. Notably,

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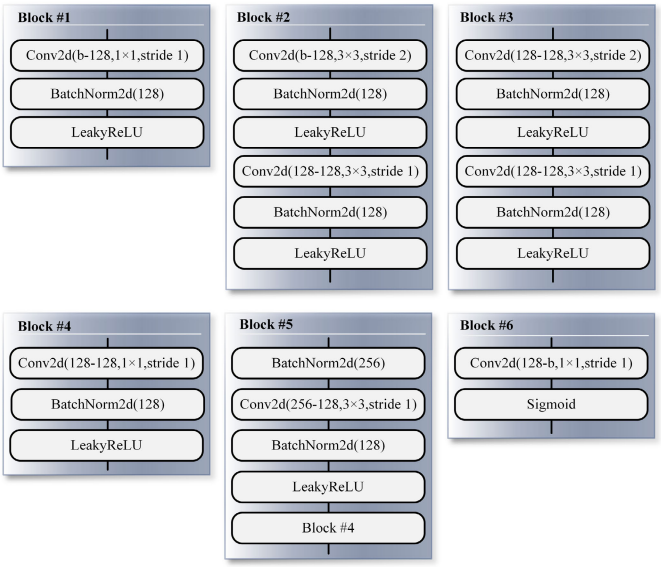


Fig. 4. Detailed architecture of the proposed network blocks.

each training iteration of the network corresponds to one step of the ADMM algorithm, establishing a principled link between deep learning and model-based optimization.

C. Joint Low-Rank and Sparse Priors Regularized Fully Convolutional AE

The framework employed in this article is a purely convolutional AE with five shortcut connections, as shown in Fig. 3. Shortcut connections can reintroduce fine-grained spatial information from early encoder stages [58], thereby preserving spatial details in the reconstructed background. A detailed view of the convolutional AE blocks is provided in Fig. 4.

The encoder contains 15 convolutional layers, with batch normalization and LeakyReLU activation applied after each layer. The input HSI, with b spectral bands, is simultaneously fed into Block #1 and Block #2. Block #1 comprises a single 1×1 convolutional layer with stride 1, which projects the b-dimensional input into a 128-D feature space; its output is not passed to subsequent layers but is instead retained for a shortcut connection and later concatenated with corresponding features in the decoder. Block #2 contains two 3×3 convolutional layers: the first uses a stride of two to perform $2\times$ spatial downsampling while mapping the b-dimensional input to 128 channels, and the second (with stride 1) further refines the extracted features. This downsampling strategy helps progressively suppress the influence of anomalous targets, which are typically small in spatial extent and occupy only a minor fraction of the total pixels. The output of Block #2 is then directed to both Block #3 and Block #4. Blocks #3 and #4 replicate the structures of Blocks #2 and #1, respectively, with the key difference that their inputs are already 128-D.

The decoder, comprising 11 convolutional layers, follows a mirror-symmetric architecture relative to the encoder. It upsamples features via nearest-neighbor interpolation and merges them with the corresponding encoder outputs through shortcut connections, resulting in 256-D features that serve as input to Block #5. Block #5 comprises two 1×1 convolutional layers with stride 1, where the first reduces the channel dimensionality from 256 back to 128. Finally, to align the output with the input HSI in both space and spectrum, Block #6 employs a 1×1 convolution coupled with a sigmoid function, thereby reconstructing the background component.

Furthermore, to further suppress the influence of potential anomalies during training, we introduce an adaptive weighting mechanism into the loss function

$$\mathcal{L} = \frac{\rho}{2}\left(\left\|\mathbf{W}\odot\left(\mathbf{Y}-f_\theta(\mathbf{Z})-\mathbf{E}^{(k)}-\frac{\mathbf{U}_1^{(k)}}{\rho}\right)\right\|_F^2 + \left\|\mathbf{W}\odot\left(f_\theta(\mathbf{Z})-\mathbf{X}^{(k+1)}-\frac{\mathbf{U}_2^{(k)}}{\rho}\right)\right\|_F^2\right) \tag{20}$$

where $\mathbf{W} \in \mathbb{R}^{w\times h}$ is the adaptive-weighted map. $\odot$ denotes the element wise multiplication, and $\mathbf{W}$ is broadcast to all channels. To produce the adaptive-weighted map $\mathbf{W}$, the residual HSI $\mathbf{R}^{(k)} = \{\mathbf{r}_i^{(k)} \in \mathbb{R}^b\}_{i=1}^{w\times h}$ is first computed

$$\mathbf{R}^{(k)} = \mathbf{Y} - f_{\theta^{(k)}}(\mathbf{Z}). \tag{21}$$

Then, the expression for $\mathbf{W}$ is given below

$$\mathbf{w}_i = \max_{j=1}^{w\times h}\left\{\left\|\mathbf{r}_j^{(k)}\right\|_2\right\} - \left\|\mathbf{r}_i^{(k)}\right\|_2, \quad i = 1, 2, \ldots, w\times h \tag{22}$$

where $\|\mathbf{r}_i^{(k)}\|_2$ denotes the reconstruction error of the ith pixel, and $\mathbf{w}_i$ denotes the ith element of $\mathbf{W}$. The values of $\mathbf{W}$ are normalized to the range [0, 1].

Since the reconstruction error is typically higher for anomalous pixels compared to background pixels, $\mathbf{W}$ has lower values at anomalous pixels and higher values at background pixels. This implies that anomalous pixels receive insufficient learning emphasis during training, as the model is biased toward reconstructing background content. Moreover, $\mathbf{W}$ is adaptively updated during the iterative process.

D. Anomaly Detection

We utilize the reconstruction error to achieve anomaly detection. At the end of the iterative optimization, the trained network will output the reconstructed background $f_{\theta^{(N)}}(\mathbf{Z})$. The discrepancy between $\mathbf{Y}$ and $f_{\theta^{(N)}}(\mathbf{Z})$ serves as a direct indicator of the anomaly score. Then, the residual HSI $\mathbf{R} = \{\mathbf{r}_i \in \mathbb{R}^b\}_{i=1}^{w\times h}$ is first computed as

$$\mathbf{R} = \mathbf{Y} - f_{\theta^{(N)}}(\mathbf{Z}). \tag{23}$$

The l_2-norm is employed to quantify perpixel anomaly magnitude in the HSI

$$D(\mathbf{r}_i) = \|\mathbf{r}_i\|_2 \tag{24}$$

where $\mathbf{r}_i$ denotes the ith pixel of $\mathbf{R}$. The indicator $D(\mathbf{r}_i)$ is then assembled into the final detection map $\mathbf{D} = \{D(\mathbf{r}_i) \in \mathbb{R}\}_{i=1}^{w\times h}$.

In the implementation, the HSI $\mathbf{Y}$ should be normalized to the range [0, 1]. The network input $\mathbf{Z}$ is a noise-only image whose values are uniformly distributed in the interval [0,0.1] and drawn independently. The network parameters are updated

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using the Adam backpropagation algorithm [59], and the learning rate is set to 0.01. $\mathbf{X}^{(0)}, \mathbf{E}^{(0)}, \mathbf{U}_1^{(0)}, \mathbf{U}_2^{(0)}$ are initialized as zero tensors at the beginning of the iteration, and the penalty parameters λ and ρ are set to 0.5 and 1, respectively.

Algorithm 1 Joint Low-Rank and Sparse AE

Input: HSI $\mathbf{Y} \in \mathbb{R}^{b\times(w\times h)}$

Initialization: uniform noise image $\mathbf{Z} \in \mathbb{R}^{b\times(w\times h)}, \rho = 1, \lambda = 0.5, \mathbf{X}^{(0)} = \mathbf{E}^{(0)} = \mathbf{U}_1^{(0)} = \mathbf{U}_2^{(0)} = 0 \in \mathbb{R}^{b\times(w\times h)}, k = 0, iter_{\max} = 150, \mathbf{W} = 0 \in \mathbb{R}^{w\times h}$.

While $k < iter_{\max}$, **do:**

1: Update the low-rank background $\mathbf{X}^{(k+1)}$ using (13)-(15);
2: Update the weight map $\mathbf{W}$ using (21)-(22);
3: Update the network parameter $\theta^{(k+1)}$ using (20);
4: Update the sparse anomaly $\mathbf{E}^{(k+1)}$ using (17)-(18);
5: Update the Lagrange multiplier $\mathbf{U}_1^{(k+1)}, \mathbf{U}_2^{(k+1)}$ using (19);
6: $k = k + 1$

End

Obtain the trained network parameters θ and reconstructed background $f_\theta(\mathbf{Z})$.

Output: Anomaly detection map $\mathbf{D} = \|\mathbf{Y} - f_\theta(\mathbf{Z})\|_2$.

Algorithm 1 provides a complete overview of the proposed JLSAE method. To provide a visual insight into the iterative process, Fig. 5 depicts the change of the low-rank part $\mathbf{X}$, the network output $f_\theta(\mathbf{Z})$, the sparse part $\mathbf{E}$, and the detection map $\mathbf{D}$ over the HYDICE dataset under different iterations. The $AUC_{(D,F)}$ values (which will be introduced in Section IV-C) corresponding to the detection maps at each iteration are annotated at the bottom. As iterations proceed, the network output $f_\theta(\mathbf{Z})$ gradually captures the dominant structured patterns of the background, becoming smoother and more coherent, which signifies its implicit convergence toward the underlying low-dimensional manifold. Meanwhile, the sparse part $\mathbf{E}$ effectively isolates the anomalies, with its nonzero entries concentrating on anomalous regions. Crucially, the detection response maps $\mathbf{D}$ and the corresponding $AUC_{(D,F)}$ values illustrate the optimization trajectory from failure to success. In the initial stages (e.g., the 30th iteration), the response map exhibits significant background noise and false alarms, resulting in a relatively low $AUC_{(D,F)}$ of 0.7923. However, as the optimization progresses, these false alarms are effectively suppressed while the anomaly signals are enhanced. This is evidenced by the progressively cleaner background in $\mathbf{D}$ and the monotonic increase in $AUC_{(D,F)}$, which reaches 0.9996 by the 150th iteration.

IV. Experiments and Analyses

The proposed JLSAE method is evaluated on four standard HSI datasets to validate its performance in anomaly detection. These datasets cover a variety of scenes and include diverse anomalous targets like storage boxes, vehicles, buildings, etc. We compare the proposed JLSAE against several HAD methods, including statistical-based methods GRX [9], traditional low-rank and sparsity-based methods (LRASR [18], LRaSMD [19], LSMAD [21], LSDM-MoG [22]), and deep learning-based methods (RGAE [60], SSHAD [37], GT-HAD [32], MFTNet [61], Auto-AD [29], DeepLR [41], DLRSPs-DAEs [45]). Meanwhile, parameter analysis was performed on the regularization coefficients ρ and λ, and ablation experiments were performed to evaluate the contribution of key modules.

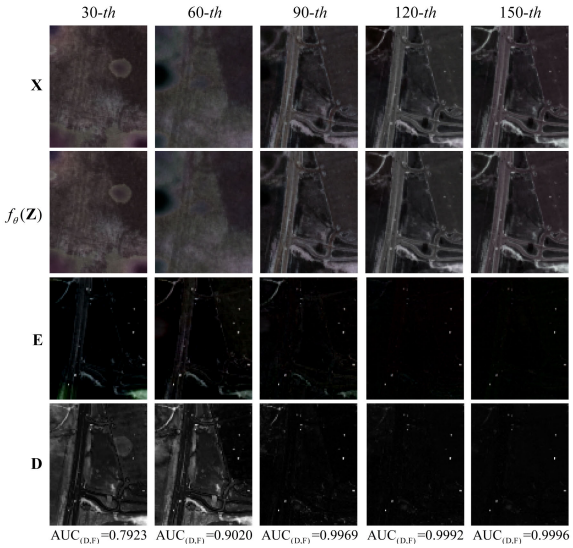


Fig. 5. Visualization of the optimization process over the HYDICE dataset under different iterations.

A. Experimental Datasets

The details of the four public datasets used are as follows:

1) *HYDICE:*[1] The first dataset was collected by the HYDICE sensor above a suburban region of Michigan, USA. This dataset comprises a 100×80 pixel grid and 162 spectral bands, with a spectral bandwidth of 10 nm and a ground sampling distance of 1 m. Several vehicles located in the scene serve as anomalies, covering 19 pixels.
2) *Hyperion:*[2] The second dataset was collected by the EO-1 Hyperion sensor over a farm region in Indiana. This dataset comprises a 150×150 pixel grid, corresponding to a ground sampling distance of 10 m and a spectral interval of 10 nm. After discarding noise-contaminated bands, 155 spectral bands were preserved. Several storage boxes and rooftops are identified as anomalies, occupying a total of 15 pixels.
3) *Bay Champagne:*[3] The third dataset originates from observations captured by the AVIRIS sensor in July 2010 over the Bay Champagne. The image size is 100×100 pixels with a spatial resolution of 4.4 m, and it contains 188-spectral bands with a 10-nm spectral interval. 11 pixels in the upper left region are identified as anomalies.
4) *Pavia:*[2] The fourth dataset originates from observations captured by the ROSIS-03 sensor in the Pavia region. The image size is 100×100 pixels, corresponding to a ground sampling distance of 1.3 m, and provides spectral measurements at a 4-nm interval. After removing bands

[1]https://github.com/xdjiangkai/LREN/tree/main/datasets/avirs
[2]https://github.com/lzw-lzw/Hyperspectral-anomaly-detectionmethods/tree/master/DATASET
[3]http://xudongkang.weebly.com/data-sets.html

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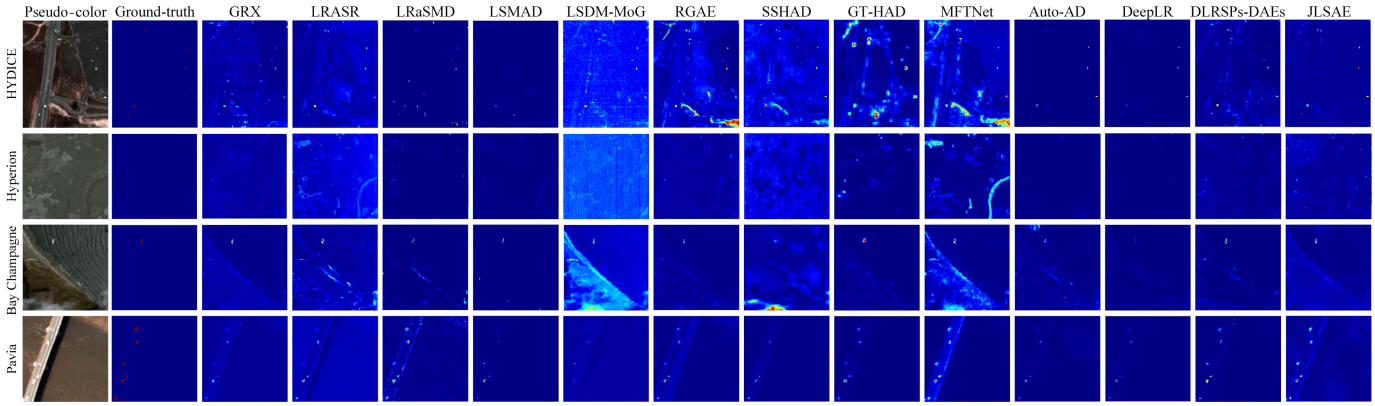


Fig. 6. Pseudo-color images, ground truth, and detection results of different comparison methods on four datasets.

corrupted by noise, 102 clean spectral bands remain. Anomalous targets correspond to the vehicles on the bridge, occupying 71 pixels. Fig. 6 displays the ground-truth maps together with their associated pseudocolor images for the datasets introduced above.

B. Implementation Details

To ensure the fairness of the experiment, the parameters of all methods were optimized on the datasets used. For the LRASR method, the number of clusters K, the number of atoms selected per cluster P, and the regularization parameters β and λ adopted on the HYDICE, Bay Champagne, Hyperion, and Pavia datasets were (25, 20, 0.01, 0.1), (10, 20, 0.1, 0.1), (20, 30, 0.1, 0.1), and (25, 20, 0.1, 0.1), respectively. For the LRaSMD method, the rank r and sparsity k were optimized to (4, 0.5), (8, 0.1), (7, 0.7), and (7, 0.7) on the four datasets, respectively. For the LSMAD method, the rank r and sparsity k were optimized to (5, 0.1), (8, 0.1), (4, 0.3), and (4, 0.1) on the four datasets, respectively. For the LSDM-MoG method, it achieves the best detection results when the initial rank is set to $0.6 \times b$ (where b is the number of spectral bands) and the Gaussian mixture size K is set to 4. For the RGAE method, on the four datasets, the superpixel quantity S and the hidden layer dimension n_{hid} were set to 150 and 100, respectively. The tradeoff parameter λ was set to 0.001 on Bay Champagne and 0.01 on the other three datasets. For the SSHAD method, the loss weight λ was set to 0.1 on HYDICE and 0.01 on the remaining three datasets. The epoch, batch size, and learning rate of the GT-HAD method were set to 150, 64, and 0.001, respectively, using the Adam optimizer. For the MFTNet method, the number of hidden layer nodes was 16, the number of attention heads was 1, the fusion weight was 0.5, and it was trained with the Adam optimizer and a learning rate of 0.0001. For the Auto-AD and DeepLR methods, the learning rate was set to 0.01, using the Adam optimizer. The parameters of DLRSPs-DAEs include the hidden nodes count n_{hid}, the tradeoff parameters $\lambda, \beta, \gamma, \alpha$, and ν, the learning rate lr, and the transformation coefficient k, which were set to 20, 0.001, 0.01, 0.01, 0.1, 0.01, 0.0001, and 0.001, respectively, using the Adam optimizer.

The comparative methods selected in the experiments include both classical signal modeling and modern neural architectures. Most of them are either based on low-rank and sparse decomposition (e.g., LRASR, LRaSMD, LSMAD, LSDM-MoG) or AE architectures (e.g., RGAE, DLRSPs-DAEs, Auto-AD, DeepLR), which are highly relevant to our JLSAE method. Therefore, the comparative experiments can effectively demonstrate the superiority and advancement of the JLSAE within this closely related family of approaches. Meanwhile, due to the randomness in the initialization of the aforementioned methods (except for GRX), the results vary across different runs. Therefore, we conducted five independent runs and calculated the mean and standard deviation of the evaluation metrics.

C. Evaluation Metrics

To overcome the limitations of traditional 2-D ROC curves in the evaluation of HAD, this article adopts a 3-D ROC curve analysis method based on the threshold parameter τ [62], [63], [64]. Unlike traditional methods that only plot the relationship between detection probability P_D and false alarm probability P_F, the 3-D ROC curve treats the threshold τ as an independent variable, constructing curves composed of three dimensions: $(P_D, P_F), (P_D, \tau)$, and (P_F, τ). Based on the 3-D ROC curve, eight metrics are generated: $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{\mathrm{ADP}}$, $\mathrm{AUC}_{\mathrm{BDP}}$, $\mathrm{AUC}_{\mathrm{JAD}}$, $\mathrm{AUC}_{\mathrm{JBS}}$, $\mathrm{AUC}_{\mathrm{ADBS}}$, $\mathrm{AUC}_{\mathrm{SNPR}}$, and $\mathrm{AUC}_{\mathrm{OADP}}$

$$
\begin{aligned}
&0 \le \mathrm{AUC}_{(\mathrm{D,F})} \le 1\\
&0 \le \mathrm{AUC}_{\mathrm{ADP}} = \mathrm{AUC}_{(\mathrm{D},\tau)} \le 1\\
&0 \le \mathrm{AUC}_{\mathrm{BDP}} = 1 - \mathrm{AUC}_{(\mathrm{F},\tau)} \le 1\\
&0 \le \mathrm{AUC}_{\mathrm{JAD}} = \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{\mathrm{ADP}} \le 2\\
&0 \le \mathrm{AUC}_{\mathrm{JBS}} = \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{\mathrm{BDP}} \le 2\\
&0 \le \mathrm{AUC}_{\mathrm{ADBS}} = \mathrm{AUC}_{\mathrm{ADP}} + \mathrm{AUC}_{\mathrm{BDP}} \le 2\\
&0 < \mathrm{AUC}_{\mathrm{SNPR}} = \frac{\mathrm{AUC}_{(\mathrm{D},\tau)}}{\mathrm{AUC}_{(\mathrm{F},\tau)}}\\
&0 \le \mathrm{AUC}_{\mathrm{OADP}} = \mathrm{AUC}_{(\mathrm{D,F})}\\
&\qquad + \mathrm{AUC}_{\mathrm{ADP}} + \mathrm{AUC}_{\mathrm{BDP}} \le 3
\end{aligned}
\tag{25}
$$

where $\mathrm{AUC}_{(\mathrm{D,F})}$ denotes the area under the 2-D ROC curve (P_D, P_F), $\mathrm{AUC}_{(\mathrm{D},\tau)}$ denotes the area under the 2-D ROC curve (P_D, τ), and $\mathrm{AUC}_{(\mathrm{F},\tau)}$ denotes the area

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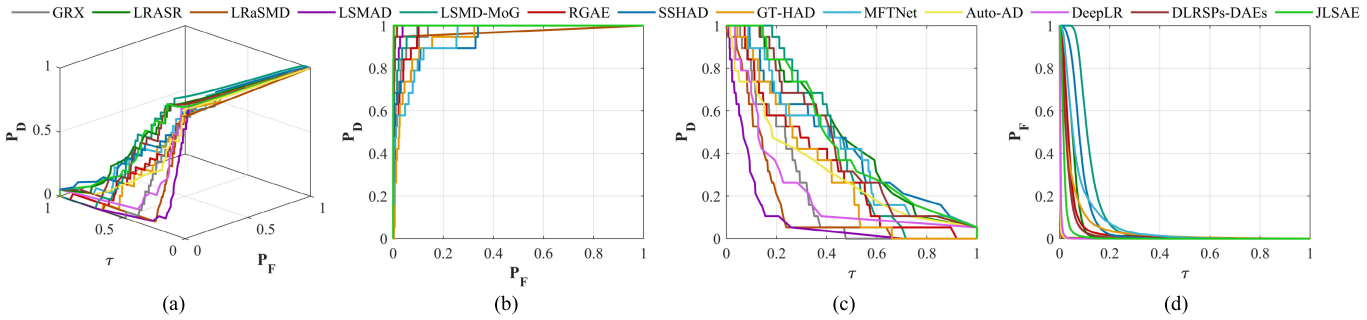


Fig. 7. ROC curves on the HYDICE dataset. (a) 3-D ROC curve. (b) 2-D ROC curve (P_D, P_F). (c) 2-D ROC curve (P_D, τ). (d) 2-D ROC curve (P_F, τ).

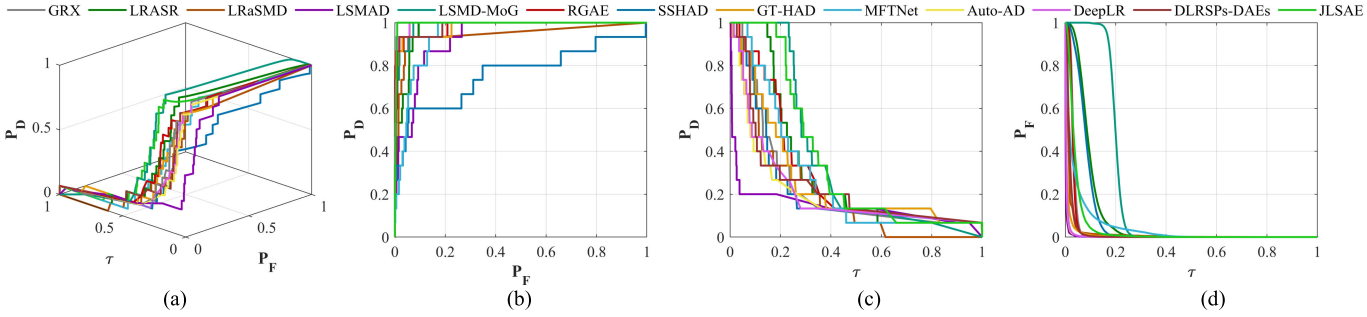


Fig. 8. ROC curves on the Hyperion dataset. (a) 3-D ROC curve. (b) 2-D ROC curve (P_D, P_F). (c) 2-D ROC curve (P_D, τ). (d) 2-D ROC curve (P_F, τ).

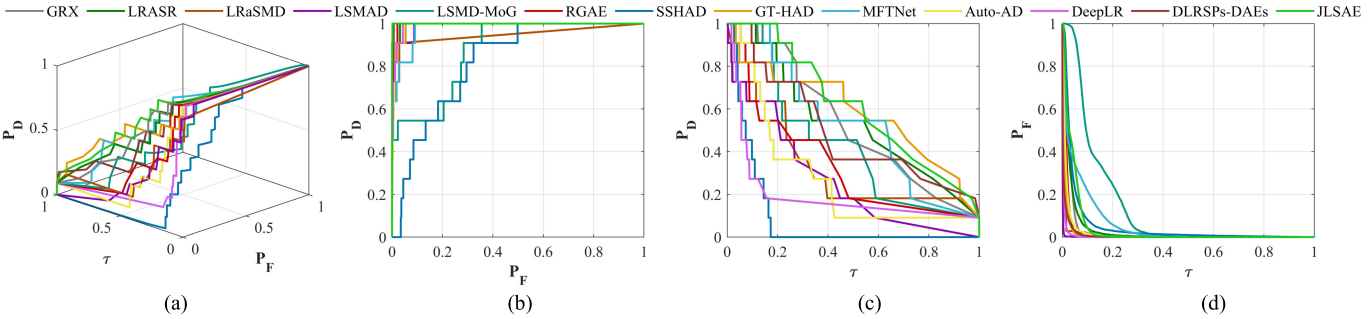


Fig. 9. ROC curves on the Bay Champagne dataset. (a) 3-D ROC curve. (b) 2-D ROC curve (P_D, P_F). (c) 2-D ROC curve (P_D, τ). (d) 2-D ROC curve (P_F, τ).

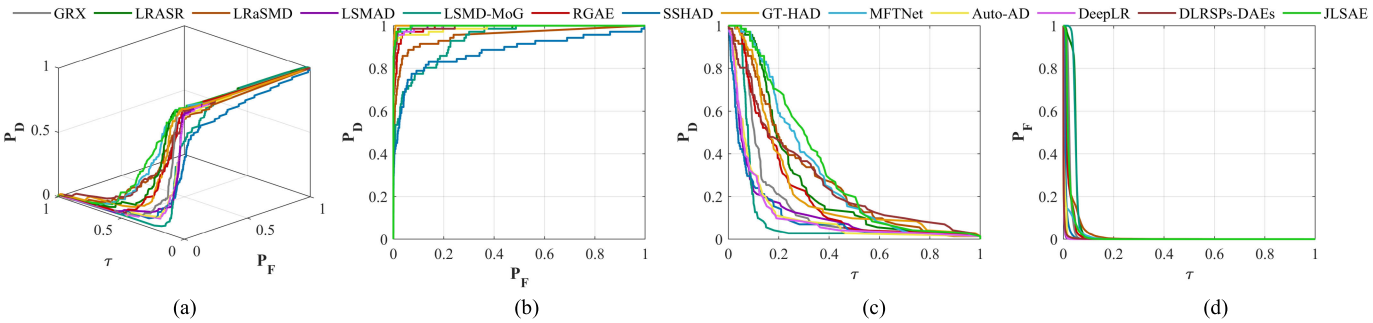


Fig. 10. ROC curves on the Pavia dataset. (a) 3-D ROC curve. (b) 2-D ROC curve (P_D, P_F). (c) 2-D ROC curve (P_D, τ). (d) 2-D ROC curve (P_F, τ).

under the 2-D ROC curve (P_F, τ). $\text{AUC}_{(\text{D,F})}$ reflects the effectiveness of a detector, AUC_{ADP} represents the anomaly detection probability, AUC_{BDP} represents the background detection probability (background suppression), AUC_{JAD} represents the joint anomaly detection, AUC_{JBS} represents the joint background suppression, AUC_{ADBS} represents the joint anomaly detection probability and background suppression, AUC_{SNPR} represents the signal-to-noise probability ratio, and AUC_{OADP} represents the overall anomaly detection probability. For all the above metrics, higher values indicate better performance. In particular, the threshold is set based on data sample values, and the AUC values are normalized to

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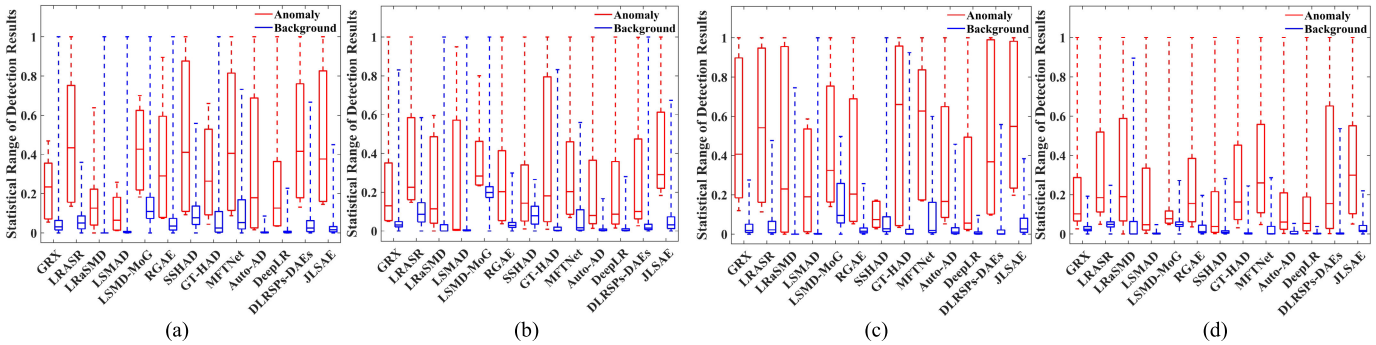


Fig. 11. Box plots of competing detectors on four hyperspectral datasets. (a) HYDICE. (b) Hyperion. (c) Bay Champagne. (d) Pavia.

TABLE I
AUC VALUES OF THE 13 COMPETING DETECTORS ON HYDICE DATASET

	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
GRX	0.9763	0.2184	0.9620	1.1947	1.9382	1.1803	5.7406	2.1566
LRASR	0.9988±0.0001	**0.4752±0.0036**	0.9473±0.0013	**1.4740±0.0037**	1.9461±0.0013	1.4225±0.0039	9.0248±0.2383	2.4214±0.0040
LRaSMD	0.9705±0.0264	0.2158±0.0591	0.9925±0.0029	1.1863±0.0627	1.9630±0.0267	1.2083±0.0563	29.8572±3.7943	2.1788±0.0601
LSMAD	0.9942±0.0014	0.1680±0.1513	0.9937±0.0060	1.1621±0.1511	1.9879±0.0066	1.1617±0.1454	30.8066±14.1358	2.1558±0.1452
LSMD-MoG	0.9709±0.0141	0.4217±0.0229	0.8725±0.0239	1.3926±0.0355	1.8434±0.0368	1.2942±0.0464	3.4577±1.0161	2.2651±0.0589
RGAE	0.9735±0.0015	0.3290±0.0110	0.9555±0.0012	1.3025±0.0095	1.9290±0.0023	1.2845±0.0104	7.4014±0.2286	2.2580±0.0089
SSHAD	0.9543±0.0083	0.4436±0.0247	0.9133±0.0160	1.3979±0.0198	1.8676±0.0222	1.3569±0.0090	5.1975±0.6725	2.3112±0.0052
GT-HAD	0.9427±0.0051	0.3008±0.0208	0.9491±0.0037	1.2436±0.0225	1.8918±0.0080	1.2499±0.0188	5.9163±0.3763	2.1926±0.0212
MFTNet	0.9455±0.0112	0.4190±0.0230	0.9206±0.0057	1.3645±0.0306	1.8661±0.0113	1.3395±0.0180	5.2800±0.1732	2.2850±0.0265
Auto-AD	0.9989±0.0008	0.3117±0.0384	**0.9969±0.0011**	1.3106±0.0379	**1.9958±0.0016**	1.3086±0.0375	**105.5421±23.3145**	2.3075±0.0370
DeepLR	0.9991±0.0006	0.2418±0.0421	0.9959±0.0007	1.2410±0.0415	1.9951±0.0013	1.2377±0.0414	59.6719±3.5134	2.2369±0.0408
DLRSPs-DAEs	0.9946±0.0014	0.4108±0.0783	0.9764±0.0050	1.4054±0.0789	1.9710±0.0037	1.3871±0.0753	17.3895±2.2082	2.3871±0.0759
JLSAE	**0.9997±0.0001**	0.4492±0.0155	0.9788±0.0024	1.4488±0.0155	1.9783±0.0024	**1.4280±0.0179**	21.4478±3.2581	**2.4276±0.0179**

address the issue of the ROC curve not passing through (0,0) or (1,1) [62]. For the MATLAB code of these AUC metrics and 3D-ROC curves, we refer to [62].[4] Moreover, box plots provide an intuitive visualization of the separability between the anomalies and the background. The anomaly response values of anomalous and background pixels are displayed in two distinct boxes, with red denoting anomalies and blue representing the background. The ends of the box plot indicate the maximum and minimum values, the upper and lower edges represent the 90th and 10th percentiles, respectively, and the middle line represents the median. Effective background suppression is indicated by a narrow background box, while the pronounced gap between the anomaly and background boxes reflects the degree of separation between anomalous and background pixels in the detection score space. The box length was empirically chosen from [10%, 90%], meaning the box plot does not represent the detection results for every threshold [65], [66]. The 3D-ROC, as a powerful indicator that accounts for different thresholds, effectively bridges this gap [62], [63], [64].

D. Comparative Evaluation of Detection Performance

1) HYDICE: As shown in the first row of Fig. 6, visually, LRaSMD, LSMAD, Auto-AD, DeepLR, DLRSPs-DAEs, and JLSAE all achieve relatively good background suppression. With the aid of quantitative metrics AUC_{BDP} and AUC_{JBS} in Table I, it can be observed that Auto-AD and DeepLR achieve the best background suppression. Auto-AD also attains the highest AUC_{SNPR} by virtue of its superior background suppression. The LSDM-MoG, RGAE, SSHAD, GT-HAD, and MFTNet exhibit obvious false alarms in their detection maps, consequently yielding lower AUC_{BDP} scores. In terms of anomaly detection probability, except for LSMAD, which shows obvious missed detections, all other methods visually detect almost all anomaly targets. However, as indicated by AUC_{ADP} and AUC_{JAD}, LRASR achieves the best anomaly enhancement effect, followed by JLSAE. Although JLSAE does not possess the highest AUC_{ADP} and AUC_{BDP}, it achieves the highest AUC_{ADBS}. This demonstrates that JLSAE has the strongest comprehensive capability in suppressing the background and detecting anomalies. Meanwhile, JLSAE also obtains the highest $AUC_{(D,F)}$ and AUC_{OADP}, indicating it has the best overall detection performance. Interestingly, compared with the DeepLR method constrained only by low-rankness, the DLRSPs-DAEs method based on low-rank and sparsity constraints shows a decrease in AUC_{BDP} but an improvement in AUC_{ADP}. This suggests that the model's focus shifts from solely attending to the low-rank property of the background to simultaneously considering the low-rank property of the background and the sparsity of the target. Under the tradeoff between suppressing background and highlighting targets, AUC_{BDP} of DLRSPs-DAEs decreases slightly (a 1.96% drop relative to DeepLR), while AUC_{ADP} improves significantly

[4]https://umbc.atlassian.net/wiki/spaces/rssipl/pages/27885869/10.+Download

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TABLE II
AUC VALUES OF THE 13 COMPETING DETECTORS ON HYPERION DATASET

	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
GRX	0.9896	0.2327	0.9669	1.2223	1.9566	1.1996	7.0354	2.1893
LRASR	0.9705±0.0014	0.3093±0.0083	0.9067±0.0008	1.2798±0.0096	1.8772±0.0011	1.2160±0.0077	3.3160±0.0703	2.1865±0.0090
LRaSMD	0.9467±0.0122	0.1610±0.0579	0.9863±0.0054	1.1077±0.0477	1.9329±0.0159	1.1473±0.0539	12.2885±4.4543	2.0940±0.0437
LSMAD	0.9553±0.0274	0.1488±0.0430	0.9947±0.0046	1.1040±0.0560	1.9500±0.0285	1.1435±0.0455	**44.0233±24.1204**	2.0987±0.0583
LSMD-MoG	0.9871±0.0039	0.3624±0.0207	0.8002±0.0052	1.3495±0.0168	1.7873±0.0091	1.1626±0.0155	1.8139±0.0544	2.1497±0.0116
RGAE	0.9822±0.0030	0.2852±0.0056	0.9698±0.0009	1.2674±0.0072	1.9521±0.0032	1.2550±0.0048	9.4528±0.1378	2.2372±0.0066
SSHAD	0.6780±0.0603	0.2158±0.0135	0.8991±0.0165	0.8937±0.0714	1.5771±0.0682	1.1149±0.0207	2.1851±0.3808	1.7928±0.0780
GT-HAD	0.9807±0.0054	0.2728±0.0077	0.9897±0.0009	1.2535±0.0092	1.9704±0.0053	1.2625±0.0068	26.5358±1.6609	2.2432±0.0083
MFTNet	0.9400±0.0097	0.2631±0.0191	0.9589±0.0076	1.2031±0.0263	1.8989±0.0146	1.2220±0.0192	6.5591±1.2128	2.1620±0.0275
Auto-AD	0.9941±0.0017	0.2015±0.0128	**0.9947±0.0015**	1.1956±0.0117	1.9887±0.0030	1.1962±0.0114	40.8367±13.6479	2.1902±0.0104
DeepLR	0.9951±0.0023	0.2056±0.0057	0.9945±0.0008	1.2006±0.0067	**1.9896±0.0023**	1.2001±0.0052	38.1977±4.4898	2.1952±0.0062
DLRSPs-DAEs	0.9870±0.0041	0.2342±0.0402	0.9749±0.0042	1.2212±0.0386	1.9619±0.0033	1.2092±0.0392	9.5370±2.1392	2.1962±0.0373
JLSAE	**0.9954±0.0007**	**0.3656±0.0124**	0.9629±0.0054	**1.3609±0.0122**	1.9582±0.0054	**1.3285±0.0095**	9.9846±1.2174	**2.3238±0.0093**

TABLE III
AUC VALUES OF THE 13 COMPETING DETECTORS ON BAY CHAMPAGNE DATASET

	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
GRX	**0.9998**	0.5314	0.9741	1.5313	1.9739	1.5055	20.4912	2.5053
LRASR	0.9949±0.0006	0.5301±0.0097	0.9628±0.0086	1.5250±0.0093	1.9577±0.0091	1.4929±0.0078	14.8192±3.0991	2.4878±0.0078
LRaSMD	0.9334±0.0469	0.3431±0.1574	0.9801±0.0080	1.2765±0.1380	1.9135±0.0504	1.3232±0.1537	18.9185±8.5463	2.2566±0.1348
LSMAD	0.9916±0.0054	0.2665±0.0750	**0.9967±0.0032**	1.2580±0.0776	1.9883±0.0069	1.2632±0.0748	**119.8239±63.2863**	2.2547±0.0774
LSMD-MoG	0.8801±0.0062	0.4437±0.0538	0.8765±0.0362	1.3238±0.0594	1.7566±0.0322	1.3202±0.0229	3.7558±0.7287	2.2003±0.0290
RGAE	0.9960±0.0006	0.3435±0.0250	0.9797±0.0081	1.3396±0.0249	1.9758±0.0083	1.3232±0.0170	18.2013±3.9926	2.3193±0.0169
SSHAD	0.8205±0.0375	0.0903±0.0141	0.9520±0.0045	0.9108±0.0490	1.7725±0.0354	1.0422±0.0099	1.8725±0.1382	1.8628±0.0460
GT-HAD	0.9976±0.0030	0.5773±0.0270	0.9903±0.0013	1.5748±0.0265	1.9879±0.0025	1.5676±0.0270	60.5193±8.1782	2.5652±0.0264
MFTNet	0.9713±0.0218	0.5213±0.0246	0.9413±0.0099	1.4926±0.0376	1.9126±0.0306	1.4626±0.0244	9.0919±1.5700	2.4339±0.0420
Auto-AD	0.9908±0.0020	0.2807±0.0471	0.9855±0.0043	1.2715±0.0478	1.9763±0.0051	1.2662±0.0437	20.7314±6.3088	2.2570±0.0445
DeepLR	0.9926±0.0022	0.2064±0.0118	0.9944±0.0009	1.1990±0.0133	1.9870±0.0022	1.2008±0.0110	36.9982±3.5564	2.1934±0.0126
DLRSPs-DAEs	0.9967±0.0051	0.4987±0.0473	0.9928±0.0008	1.4954±0.0464	**1.9894±0.0053**	1.4915±0.0475	69.6227±10.5033	2.4882±0.0466
JLSAE	0.9997±0.0001	**0.6120±0.0118**	0.9611±0.0043	**1.6117±0.0118**	1.9608±0.0044	**1.5730±0.0075**	15.7168±1.7065	**2.5728±0.0075**

(a 69.9% increase relative to DeepLR). Therefore, DLRSPs-DAEs achieves a higher AUC_{ADBS}. Based on maintaining the AUC_{BDP} of DLRSPs-DAEs, JLSAE further improves AUC_{ADP}, indicating that the unified framework used by JLSAE can achieve better joint optimization than the dual-branch AE framework used by DLRSPs-DAEs. From the (P_D, P_F) curve in Fig. 7, it can be seen that the curves of all methods are close to the top-left corner, indicating that all methods achieve effective detection results on the HYDICE dataset, but JLSAE is closer to the top-left corner, suggesting it is more effective. The (P_D, τ) curve indicates that the anomaly detection probabilities of LRASR, SSHAD, and the proposed JLSAE method are comparable. Although the proposed method does not have the lowest false alarm rate in the (P_F, τ) curve, it is relatively close to the bottom-left, which can be considered an upper middle level. Therefore, the advantage of the proposed JLSAE is that it can significantly improve the anomaly detection rate on the premise of ensuring the false alarm rate does not increase significantly (i.e., a substantial increase in AUC_{ADP} without a significant decrease in AUC_{BDP}). The box plot in Fig. 11(a) also shows that the proposed method has the largest separation between the background and anomaly boxes, indicating the best separation between background and anomalies is achieved.

2) Hyperion: As shown in the second row of Fig. 6, from a visual perspective, LRaSMD, LSMAD, Auto-AD, and DeepLR achieve the best background suppression performance. This is also reflected in the quantitative metrics AUC_{BDP} and AUC_{JBS} in Table II, where their values are higher than those of other methods. However, since the targets in this dataset are all very small, the anomaly responses in the detection maps of these methods are not very distinct. Although the proposed JLSAE does not achieve the highest AUC_{BDP} and AUC_{JBS}, it still maintains a competitive performance. The proposed method attains the highest AUC_{ADP} and AUC_{JAD}, indicating a significant advantage in anomaly enhancement. While LSDM-MoG achieves a competitive anomaly detection probability AUC_{ADP}, its detection map reveals severe false alarms, leading to a low AUC_{BDP} and consequently a suboptimal joint anomaly detection and background suppression score AUC_{ADBS}. The proposed JLSAE obtains the highest AUC_{ADBS}, demonstrating the strongest comprehensive capability in suppressing background and highlighting anomalies. In other words, its advantage in anomaly detection outweighs its slight inferiority in background suppression. Although Auto-AD and DeepLR yield high AUC_{BDP}, their AUC_{ADP} is not particularly high, suggesting that under conditions of excessive background suppression, the anomaly targets are also somewhat suppressed. This is likely because they do not explicitly consider anomaly target characteristics during optimization, pursuing only background suppression. The phenomenon observed on the HYDICE dataset is also observable on Hyperion: compared

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TABLE IV
AUC VALUES OF THE 13 COMPETING DETECTORS ON PAVIA DATASET

	$AUC_{(D,F)}$	AUC_{ADP}	AUC_{BDP}	AUC_{JAD}	AUC_{JBS}	AUC_{ADBS}	AUC_{SNPR}	AUC_{OADP}
GRX	0.9934	0.1600	0.9742	1.1535	1.9677	1.1343	6.2103	2.1277
LRASR	0.9909±0.0049	0.2595±0.0071	0.9506±0.0018	1.2504±0.0043	1.9415±0.0066	1.2102±0.0056	5.2590±0.0916	2.2010±0.0041
LRaSMD	0.9672±0.0151	0.2945±0.0394	0.9769±0.0028	1.2616±0.0491	1.9432±0.0161	1.2705±0.0376	12.3111±1.3970	2.2377±0.0479
LSMAD	0.9968±0.0014	0.1178±0.0227	**0.9985±0.0004**	1.1146±0.0229	1.9952±0.0014	1.1163±0.0223	78.6413±7.4032	2.1131±0.0225
LSMD-MoG	0.9388±0.0118	0.1035±0.0268	0.9550±0.0209	1.0423±0.0324	1.8938±0.0260	1.0585±0.0132	2.7510±1.2792	1.9973±0.0246
RGAE	0.9911±0.0004	0.2053±0.0041	0.9812±0.0003	1.1964±0.0045	1.9723±0.0002	1.1865±0.0039	10.9031±0.1202	2.1776±0.0042
SSHAD	0.8858±0.0145	0.1079±0.0023	0.9901±0.0002	0.9937±0.0157	1.8760±0.0146	1.0980±0.0022	10.9219±0.1450	1.9838±0.0157
GT-HAD	**0.9992±0.0001**	0.2372±0.0027	0.9980±0.0001	1.2364±0.0027	**1.9972±0.0002**	1.2351±0.0028	**117.2387±8.3376**	2.2343±0.0028
MFTNet	0.9962±0.0006	0.2997±0.0107	0.9921±0.0022	1.2959±0.0106	1.9883±0.0022	1.2917±0.0087	40.5799±13.1768	2.2880±0.0085
Auto-AD	0.9885±0.0041	0.1143±0.0146	0.9964±0.0015	1.1028±0.0111	1.9849±0.0055	1.1107±0.0132	36.0663±12.2236	2.0992±0.0097
DeepLR	0.9946±0.0016	0.1240±0.0261	0.9971±0.0018	1.1186±0.0258	1.9971±0.0025	1.1211±0.0243	49.4068±12.8276	2.1157±0.0240
DLRSPs-DAEs	0.9954±0.0010	0.2863±0.0294	0.9975±0.0002	1.2817±0.0302	1.9930±0.0010	1.2838±0.0295	116.6937±17.1334	2.2793±0.0303
JLSAE	0.9976±0.0004	**0.3229±0.0070**	0.9771±0.0020	**1.3206±0.0072**	1.9747±0.0016	**1.3000±0.0063**	14.1693±1.1430	**2.2976±0.0064**

with the DeepLR method, DLRSPs-DAEs shows a decrease in AUC_{BDP} but an improvement in AUC_{ADP}. This indicates that simultaneously considering low-rank and sparsity constraints may slightly impair the model's background suppression capability to a certain extent but significantly boosts the anomaly detection probability. The JLSAE further improves AUC_{ADP} while maintaining the AUC_{BDP} of DLRSPs-DAEs without significant change, thereby achieving the highest AUC_{ADBS}. This demonstrates that the unified framework used by JLSAE indeed achieves better joint optimization than the dual-branch AE framework used by DLRSPs-DAEs. From the (P_D, P_F) curve in Fig. 8, it can be seen that JLSAE is closest to the top-left corner, indicating the best detection efficiency. The (P_D, τ) curve is consistent with AUC_{ADP}, where both LSMD-MoG and JLSAE achieve good anomaly detection probabilities. However, the (P_F, τ) curve reveals that LSMD-MoG has a high false alarm rate, whereas JLSAE occupies an intermediate position. Therefore, the advantage of the proposed JLSAE lies in its ability to significantly improve the anomaly detection rate under the premise of a low false alarm rate. The box plot in Fig. 11(b) shows that the proposed method achieves the largest separation between the background and anomaly boxes, indicating the best separation between background and anomalies. In conclusion, JLSAE achieves the highest $AUC_{(D,F)}$ and AUC_{OADP}, demonstrating that it possesses the best overall detection performance.

3) Bay Champagne: Regarding the visualization results, as shown in the third row of Fig. 6, the proposed method can effectively suppress complex background interference while significantly enhancing the response intensity of anomaly targets. Compared with the LSMAD algorithm, although the background suppression of the proposed method is slightly inferior, the generated anomaly map exhibits better prominence in target regions, and there are no obvious false alarms in other background areas. From the quantitative indicators in Table III, LSMAD achieved the highest background detection probability AUC_{BDP}, and DLRSPs-DAEs achieved the highest AUC_{JBS}. This indicates that both methods possess good background suppression capabilities. The proposed JLSAE achieved an outstanding anomaly detection probability AUC_{ADP} (0.6120), implying a strong ability to highlight anomalies. Ultimately, on this dataset, the proposed method achieved the highest AUC_{ADBS}, demonstrating the strongest comprehensive capability for joint anomaly enhancement and background suppression, which is crucial for achieving separability between the background and anomalies. The AUC_{SNPR} of JLSAE is not the highest. This is because when calculating AUC_{SNPR}, the numerator is AUC_{ADP} and the denominator is 1-AUC_{BDP}. The order of magnitude of the denominator is very small, so a tiny change in AUC_{BDP} leads to a huge change in AUC_{SNPR}. For instance, the AUC_{ADP} of JLSAE increased by 0.4056 compared to DeepLR, while AUC_{BDP} only decreased by 0.0333. However, in the calculation of AUC_{SNPR}, the numerator becomes two times the original, but the denominator becomes seven times the original, resulting in a decrease in AUC_{SNPR}. Therefore, the most efficient way to improve AUC_{SNPR} is to improve AUC_{BDP}, an aspect where the proposed method does not have an advantage. The advantage of the proposed JLSAE is that, under the premise of maintaining a nonsignificant change in AUC_{BDP}, it significantly improves the anomaly detection probability AUC_{ADP}, thereby achieving the best comprehensive capability for enhancing anomalies and suppressing the background. From the (P_D, P_F) curve in Fig. 9, it can be seen that JLSAE is very close to the upper left corner, indicating its powerful detection capability. The (P_D, τ) curve shows that the JLSAE still maintains a relatively high anomaly detection probability compared to other methods when the threshold is close to 1. From the (P_F, τ) curve, it can be observed that the proposed method is located in an intermediate position among all methods, which is not exceptionally outstanding but still decent. From the box plot in Fig. 11(c), it can also be seen that the proposed method achieved the highest anomaly box and the largest separation between the anomaly box and the background box. LSMAD has the narrowest background box, further confirming its background suppression capability. However, its anomaly box is positioned relatively low, indicating that the anomalies were also suppressed to a certain extent. In summary, JLSAE achieved the highest overall anomaly detection probability AUC_{OADP}, indicating that it possesses the best comprehensive detection performance, with GT-HAD being the second best.

4) Pavia: The background of this dataset is relatively simple, with targets concentrated on the left side. As can be seen from the visualization in the fourth row of Fig. 6,

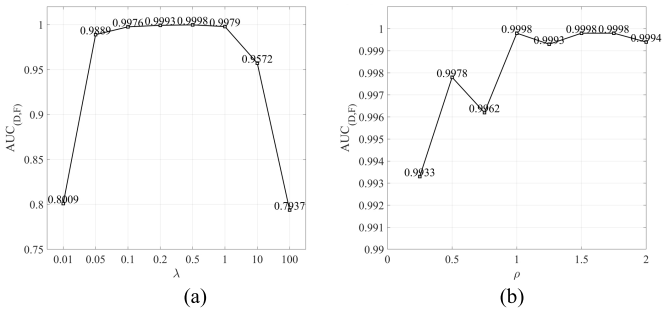
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all methods achieve relatively good background suppression outside the target aggregation area. However, near the bridge, all methods detect the contour of the bridge to varying degrees. This may be because the background is overly flat, making the bridge a relatively salient target in this scene. Nevertheless, the response of the bridge is significantly lower than that of the vehicles. Thus, selecting an appropriate threshold can still separate anomalies from the background. From the detection maps, it is evident that the anomaly responses of LRaSMD, GT-HAD, MFTNet, DLRSPs-DAEs, and JLSAE are the brightest. Combined with the AUC_{ADP} and AUC_{JAD} in Table IV, it can be concluded that JLSAE achieves the best anomaly enhancement effect, followed by MFTNet. In terms of background suppression, LSMAD and GT-HAD achieve the highest AUC_{BDP} and AUC_{JBS}, respectively. This indicates that the gating mechanism of GT-HAD indeed effectively suppresses the background. Regarding the joint capability of suppressing background while highlighting anomalies, the proposed JLSAE still achieves the highest AUC_{ADBS} on this dataset, demonstrating the stability of its anomaly-background separation capability. From the (P_D, P_F) curve in Fig. 10, it can be seen that JLSAE is close to the top-left corner, second only to GT-HAD, achieving the second-best $AUC_{(D,F)}$. The (P_D, τ) curve shows that JLSAE achieves the highest anomaly detection probability. The (P_F, τ) curve indicates that the false alarm rates of all methods are relatively low and the curves are quite close. From the box plot in Fig. 11(d), although the anomaly box of JLSAE is not as high as that of DLRSPs-DAEs, the separation between the anomaly and background boxes is much larger than that of DLRSPs-DAEs. Therefore, compared with DLRSPs-DAEs, JLSAE possesses better background and anomaly separation capability. In summary, JLSAE achieves the highest overall anomaly detection probability AUC_{OADP}, highlighting its advantage in comprehensive capability. The HYDICE and Hyperion correspond to relatively complex backgrounds, while Bay Champagne and Pavia correspond to simple backgrounds. The experimental results also demonstrate the robustness of the proposed method across scenarios of varying complexity.

E. Parameter Analysis

We further assessed the robustness and parameter sensitivity of JLSAE by analyzing its behavior with respect to two key parameters: the sparse regularization weight λ and the penalty coefficient ρ. The parameter λ governs the relative emphasis on the low-rank background and the sparse anomaly components, thereby controlling the degree of sparsity imposed on the anomaly matrix. The parameter ρ serves as the penalty factor in the augmented Lagrangian function, influencing the convergence behavior of the optimization process. To investigate their effects, each parameter was varied within a reasonable range while keeping the other fixed, and we assessed the resulting detection performance on the HYDICE dataset in terms of AUC. For brevity, AUC hereafter refers to $AUC_{(D,F)}$.

When ρ was fixed at 1, λ was varied in the range [0.01, 0.05, 0.1, 0.2, 0.5, 1, 10, 100]. As shown in Fig. 12(a), the AUC first increases rapidly as λ grows from 0.01 to 0.5, improving from 0.8009 to 0.9998. This demonstrates that a moderate sparsity strength effectively enhances anomaly discrimination. However, when λ exceeds 0.5, the AUC starts to decrease, dropping to 0.9979 at $\lambda = 1$ and further to 0.7937 when λ reaches 100. This decline occurs because overly strong sparsity regularization forces too many pixels toward zero and results in the loss of informative structure. These observations suggest that our approach is sensitive to λ and reaches its optimum when λ is set within a moderate range. Therefore, setting λ to 0.5 is recommended to achieve a favorable compromise between the low-rank and sparse constraints, thus yielding superior detection results.



(a) (b)

Fig. 12. Variation of AUC with respect to parameters on HYDICE. (a) Parameter analysis of λ. (b) Parameter analysis of ρ.

TABLE V
$AUC_{(D,F)}$ COMPARISON OF JLSAE ABLATIONS ON MULTIPLE DATASETS

Method	HYDICE	Hyperion	Bay Champagne	Pavia
w/o Sparse	0.9989	0.9917	0.9894	0.9919
w/o Low-rank	0.9986	0.9720	0.9944	0.9965
w/o L&S	0.9976	0.9812	0.9863	0.9916
w/o AE	0.9964	0.9658	0.9907	0.9938
w/o mask	0.9993	0.9819	0.9922	0.9949
full	**0.9997**	**0.9954**	**0.9997**	**0.9976**

When λ was fixed at 0.5, ρ was varied in the range [0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2]. As shown in Fig. 12(b), the AUC initially increases as ρ grows, reaching its maximum when $\rho = 1$. When ρ exceeds 1, the performance remains consistently high with only slight fluctuations, demonstrating strong robustness of the proposed method to ρ within this range. Overall, these results suggest that a moderate value of ρ, particularly around 1, leads to the best convergence behavior of the optimization process.

F. Ablation Study

We conducted the ablation study by removing key modules, including the sparse constraint (w/o Sparse), the low-rank constraint (w/o low-rank), both low-rank and sparse constraints (w/o L&S), the AE (w/o AE), and the adaptive-weighted mask (w/o mask). As shown in Table V, the full model consistently outperforms all variants in terms of AUC across every dataset.

Removing the sparse constraint results in the most significant degradation, especially on HYDICE, indicating that sparsity plays a crucial role in enhancing anomaly distinctiveness. Eliminating the low-rank constraint causes a slight yet

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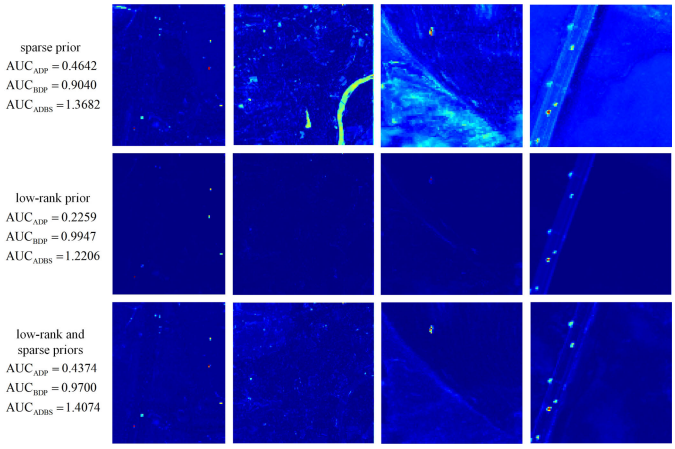


Fig. 13. Qualitative and quantitative comparison of response maps generated by sparse, low-rank, and joint low-rank sparse priors.

consistent performance drop, demonstrating that low-rankness helps model the global background structure. These findings verify that the low-rank and sparse priors are both essential, and removing either term negatively impacts the detection accuracy.

When the AE is removed, the model reduces to classical RPCA, which leads to a noticeable drop in performance and confirms the importance of its strong nonlinear feature reconstruction capability. When both low-rank and sparse constraints are removed, the performance declines further, validating the critical role of these priors in enabling the AE to accurately reconstruct the background. Meanwhile, removing the adaptive-weighted mask only causes marginal degradation, but the full model consistently achieves the best AUC.

Overall, the ablation results confirm that each module contributes to performance, and the joint use of low-rank, sparse, and AE components is crucial for robust HAD.

To better demonstrate the differences between constraints, Fig. 13 visualizes the response maps of models under sparse constraint, low-rank constraint, and joint low-rank sparse constraint across four datasets. We also calculate the average AUC_{ADP}, AUC_{BDP},and AUC_{ADBS} metrics. Visually, the model with only a sparse constraint exhibits the strongest anomaly response but suffers from significant false alarms. Conversely, the model with only a low-rank constraint achieves the best background suppression, yet the anomaly response is not distinct. The proposed joint model effectively highlights anomaly targets while maintaining a low false alarm rate. Quantitatively, a higher AUC_{ADP} indicates stronger anomaly detection capability, while a higher AUC_{BDP} signifies better background suppression. The proposed method achieves the best balance between suppressing background and highlighting anomalies, yielding the highest joint metric AUC_{ADBS}. Regarding the background suppression, it is worth noting that compared to the pure low-rank constraint, the joint constraint exhibits slightly weaker background suppression but significantly improved detection probability. This phenomenon can be attributed to the additional sparse constraint. It effectively shifts the model's focus from solely characterizing the background to a balanced modeling of both components. Consequently, while background suppression is slightly compromised, the probability of anomaly detection is significantly enhanced.

V. Conclusion

In this article, we propose JLSAE, a unified deep framework that jointly embeds low-rank and sparse priors for HAD. Unlike existing dual-branch approaches that model background and anomalies separately and rely on the decision-level fusion of their detection results, our method integrates low-rank and sparse priors within a single AE. This unified design enables end-to-end joint learning and avoids information loss caused by post hoc fusion. By interpreting training as an ADMM process, we derive a loss function with explicit low-rank and sparsity regularization, which endows the model with physically meaningful and interpretable behavior. Moreover, the proposed ADMM-inspired optimization framework is modular, allowing the AE to be replaced by other deep architectures, such as transformer-based models, while preserving the core optimization logic and extensibility. Experiments across multiple datasets demonstrate that JLSAE effectively reconstructs the background while suppressing anomalies, achieving superior detection performance compared with state-of-the-art methods. Meanwhile, we acknowledge that the decomposition framework in this work does not strictly guarantee the orthogonality between the background and anomalies. In the future, we plan to incorporate more advanced decomposition techniques into cutting-edge deep networks to further enhance background suppression and anomaly detection capabilities.

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